BIO-RAD

Gene Pulser® Electroprotocol

Cell Type Bacterial, gram positive

Species Used *Brevibacterium lactofermentum,* ATCC 13859, R31,1035 (thr-).

Molecules Electroporated DNA: covalently closed plasmids

Before the Pulse

Cell Growth Medium Tryptic soy broth (TSB)

Growth Phase at Harvest O.D. (600) =0.4 - 0.8

Pre-pulse Incubation Not given

Wash Solution Not given

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature Room temperature

Electroporation Medium 10% glycerol, distilled water

Cell Density Approx. 10 (10) cells / ml

Volume of Cells 1 to 5 µl

DNA Concentration 0.1 to 1 µg

DNA Resuspension Buffer TE buffer (10 mM Tris, 1 mM EDTA, pH8.0) or water

Volume of DNA 1 to 5 µl

Cuvette Gap 0.2 cm

Voltage 2.5 kV

Field Strength 12.5 kV/cm

Capacitor 25 µF

Resistor 200 Ω (Pulse Controller)

Time Constant 2.8 to 4.5 msec

After the Pulse

Outgrowth Medium TSB

Outgrowth Temperature 30 °C

Length of Incubation 1 hour

Selection Method or Assay Used Minimal media plus antibiotics

Electroporation Efficiency Maximum 10 (5) to10 (6) transformants / µg DNA

Per Cent Survival Not done

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

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